



American International University-Bangladesh (AIUB)

Department of Computer Science

Course Name: **COMPUTER GRAPHICS**

Faculty Name: DIPTA JUSTIN GOMES

Section: J

Student Name: **NILOY PAUL**

Course Code: CSE 4118

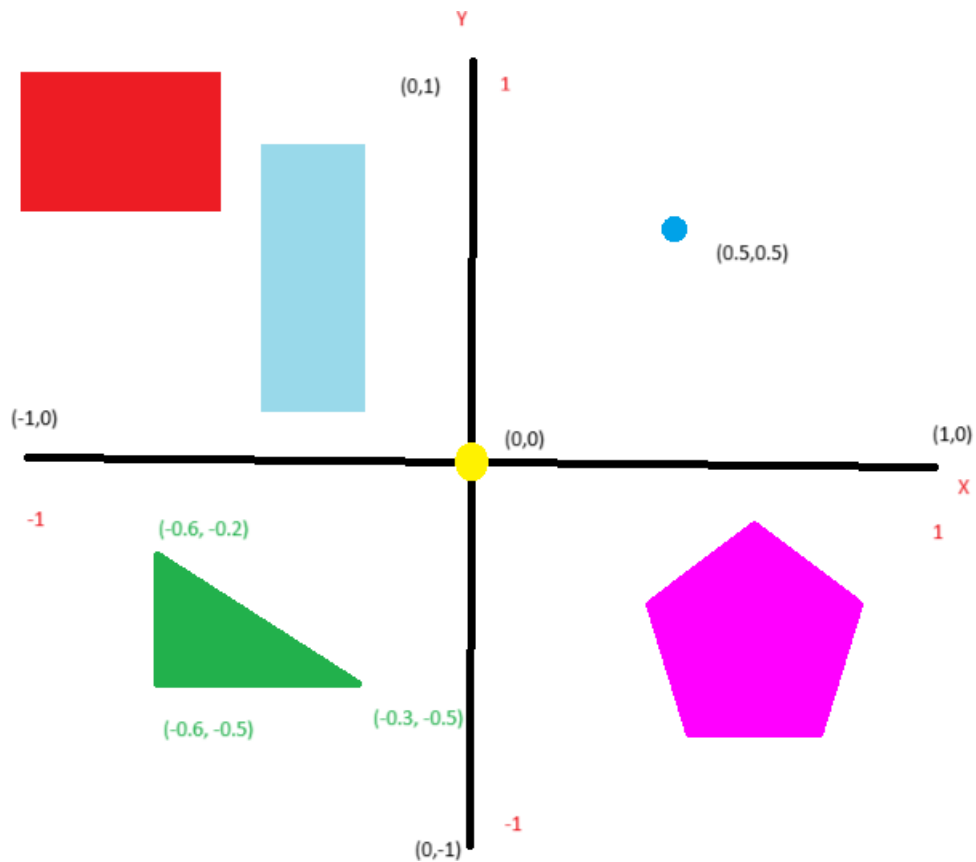
Semester: **FALL 2025-2026**

MID Term LAB Task-1

Student ID: **23-51773-2**

Basic Primitive

1. Draw the following primitives using OpenGL.



OpenGL Code

```
#include <windows.h> // for MS Windows
```

```
#include <GL/glut.h> // GLUT, include glu.h and gl.h
```

```
/* Handler for window-repaint event. Call back when the window first appears and  
whenever the window needs to be re-painted. */
```

```

void display() {
    glClearColor(1.0f, 1.0f, 1.0f, 1.0f); // Set background color to black and opaque
    glClear(GL_COLOR_BUFFER_BIT);      // Clear the color buffer (background)

    //POINT Drawing

    glPointSize(10);

    glBegin(GL_POINTS);

        //drawing the blue point in co-or (0.5,0.5)
        glColor3f(0,0,1);
        glVertex2f(0.5 ,0.5);

        //drawing the yellow point in co-or (0,0)
        glColor3f(1,1,0);
        glVertex2f(0 ,0);

    glEnd();

    //LINE Drawing

    glBegin(GL_LINES);

        glColor3f(0,0,0);

        //X-Axis
        //Drawing a line with vertex (-1,0) and (1,0)
        glVertex2f(-1,0);
        glVertex2f(1,0);

        //Y-Axis
        //drawing a line with vertex (0,-1) and (0,1)
        glVertex2f(0,-1);

```

```
glVertex2f(0,1);
```

```
glEnd();
```

```
//TRIANGLE Drawing
```

```
glBegin(GL_TRIANGLES);
```

```
glColor3f(0,1,0);
```

```
glVertex2f(-0.6, -0.2);
```

```
glVertex2f(-0.6, -0.5);
```

```
glVertex2f(-0.3, -0.5);
```

```
glEnd();
```

```
//RECTANGLE Drawing
```

```
glBegin(GL_QUADS);
```

```
glColor3f(1,0,0);
```

```
glVertex2f(-1.0, 0.9);
```

```
glVertex2f(-1.0, 0.6);
```

```
glVertex2f(-0.5, 0.6);
```

```
glVertex2f(-0.5, 0.9);
```

```
glEnd();
```

```
//RECTANGLE Drawing
```

```
glBegin(GL_QUADS);
```

```
glColor3ub(153,217,234);
```

```
glVertex2f(-0.4, 0.7);
```

```
glVertex2f(-0.4, 0.1);
```

```

    glVertex2f(-0.1, 0.1);
    glVertex2f(-0.1, 0.7);

glEnd();

//PENTAGON Drawing

glBegin(GL_POLYGON);

    glColor3f(1,0,1);
    glVertex2f(0.65, -0.1);
    glVertex2f(0.35, -0.3);
    glVertex2f(0.48, -0.65);
    glVertex2f(0.8, -0.65);
    glVertex2f(0.93, -0.3);

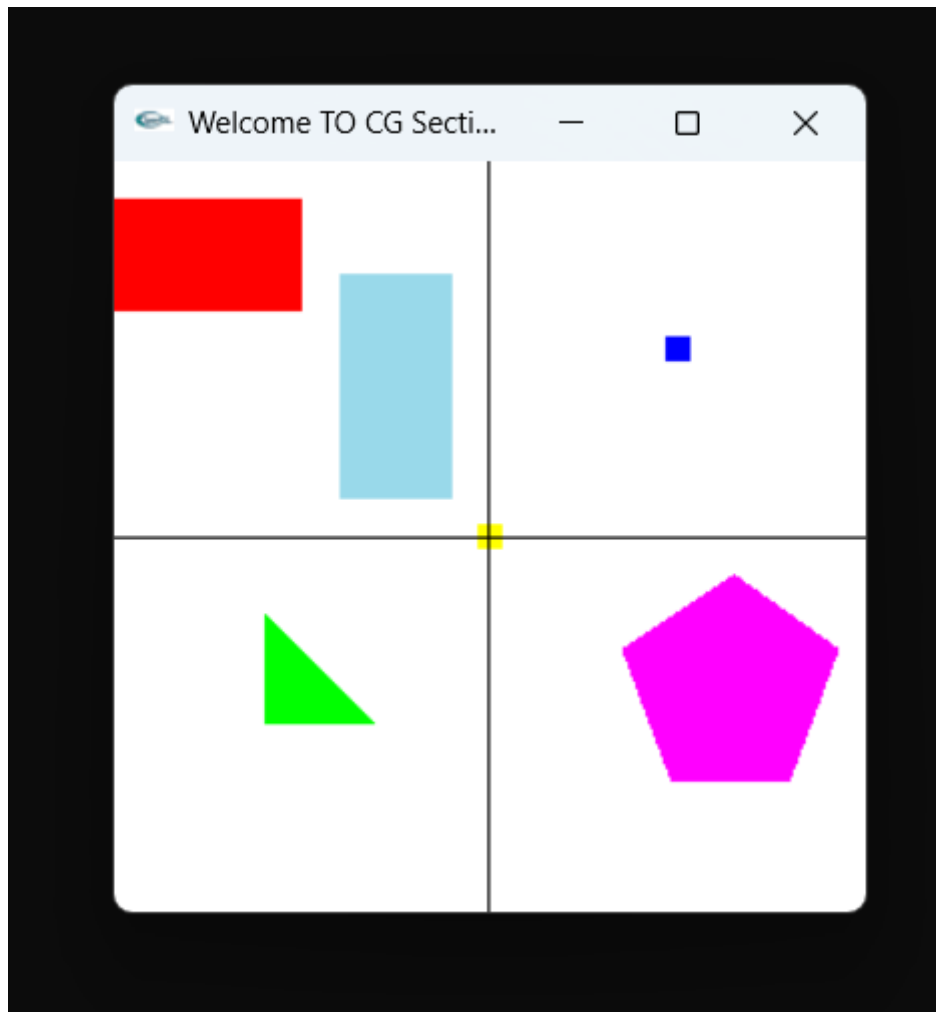
glEnd();

    glFlush(); // Render now
}

/* Main function: GLUT runs as a console application starting at main() */
int main(int argc, char** argv) {
    glutInit(&argc, argv);          // Initialize GLUT
    glutCreateWindow("Welcome TO CG Section-F"); // Create a window with the given title
    glutInitWindowSize(320, 320); // Set the window's initial width & height
    glutDisplayFunc(display); // Register display callback handler for window re-paint
    glutMainLoop();              // Enter the event-processing loop
    return 0;
}

```

Output



THE END